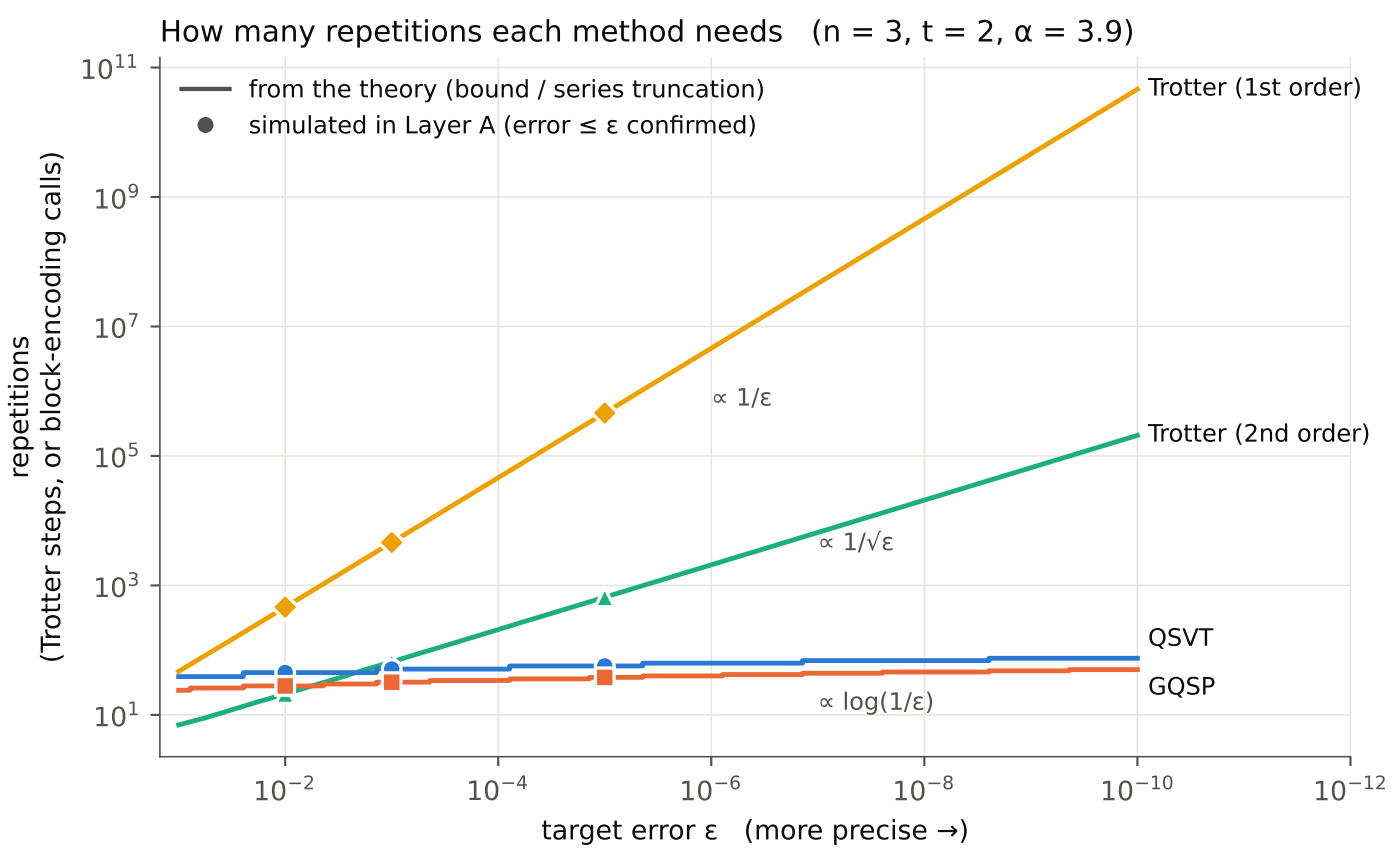
How many repetitions each method needs (n = 3, t = 2, α = 3.9)
from the theory (bound / series truncation)
simulated in Layer A (error ≤ ε confirmed)
Trotter (1st order)
Trotter (2nd order)
QSVT
GQSP
∝ 1/ε
∝ 1/√ε
∝ log(1/ε)
repetitions
(Trotter steps, or block-encoding calls)
target error ε (more precise →)
10^{11}
10^{9}
10^{7}
10^{5}
10^{3}
10^{1}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}